

Fenchel Dual

$$f: \mathbb{R}^n \rightarrow \mathbb{R} \quad \text{convex}$$

$$\nabla f: \mathbb{R}^n \rightarrow \mathbb{R} \quad \text{monotone}$$

$$\partial f: \text{set-valued mapping} \quad \text{monotone}$$

$$\text{Ex: } T: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$$T = (I + \lambda \nabla f)$$

$$\text{Gradient descent: } x \mapsto Tx$$

What about the inverse?

Inverse of the gradient?

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$\nabla f: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

For f a strongly convex quadratic: $f(x) = \frac{1}{2} x^T Q x + g^T x + c$

$$\nabla f(x) = Qx + g$$

We can compute the inverse mapping:

$$\underline{\nabla f^{-1}(y) = Q^{-1}(y - g)}$$

More generally?

Fenchel Conjugate

$$f : \mathbb{R}^n \rightarrow \mathbb{R}$$

$$f^*(y) \triangleq \sup_{x \in \text{dom } f} \{y^T x - f(x)\}$$

Fenchel's Inequality

$$f(x) + f^*(y) \geq y^T x \quad \forall x, y$$

$$\text{why: } f^*(y) = \sup_z \{y^T z - f(z)\} \geq y^T x - f(x).$$

Examples

$$f(x) = \frac{1}{2} x^T Q x + q^T x + c$$

$$f^*(y) = \sup_x \left\{ y^T x - \frac{1}{2} x^T Q x - q^T x - c \right\}$$

$$\text{Find opt } x: \quad y - Qx - q = 0$$

$$\Rightarrow x = Q^{-1}(y - q)$$

$$\Rightarrow f^*(y) = \frac{1}{2} (y - q)^T Q^{-1} (y - q)$$

$$\text{Note: } \nabla f^*(y) = Q^{-1}(y - q) = (\nabla f(x))^{-1}$$

Exercise: What if $Q \not\prec 0$ but not strictly p.d.?

$$\bullet f(x) = \sum x_i \log x_i$$

$$f^*(y) = \sum e^{y_i - 1}$$

$$f(x) = - \sum \log x_i$$

$$f^*(y) = -\sum \log(-y_i) - n$$

- $f(x) = -\log \det x$

$\uparrow$
square matrix

$$f^*(y) = -\log \det(-Y) - n$$

Even More Examples

Indicator of a convex set: $I_X(x) = \begin{cases} 0 & x \in X \\ +\infty & \text{o.w.} \end{cases}$

Def'n: Support function

$$S_X(y) \triangleq \sup_{x \in X} y^T x$$

$B_{\|\cdot\|}$ = unit ball of a norm $\|\cdot\| = \{x: \|x\| \leq 1\}$

Let $f(x) = \|x\|$

$$f^*(y) = \sup_x \{y^T x - \|x\|\} = I_{B_{\|\cdot\|^*}}$$

$$\|y\|^* = \sup_{\|x\| \leq 1} x^T y$$

If $\|y\|^* \leq 1 \Rightarrow y^T x \leq \|x\| \Rightarrow f^*(y) = 0$

$\|y\|^* > 1 \Rightarrow \exists x$ with $\|x\| \leq 1$, $x^T y > 1$, by scaling: $f^*(y) = +\infty$

Calculus Rules

Exercise: check that these hold.

- $f(x_1, x_2) = f_1(x_1) + f_2(x_2)$
 $\Rightarrow F^*(y_1, y_2) = f_1^*(y_1) + f_2^*(y_2)$
- $(a f(x))^* = a f^*(y/a)$
- $(a f(x/a))^* = a f^*(y)$

Calculus Rules - continued -

- $(f(x) + a^T x + b)^* = f^*(y - a) - b$

- $(f(x - b))^* = b^T y + f^*(y)$

- $(f(Ax))^* = f^*(A^{-T} y)$

- Infimal convolution

Given $g(x), h(x)$

$$f(x) = \inf_{u+v=x} (g(u) + h(v))$$

$$f^*(y) = g^*(y) + h^*(y).$$

$$(f(Ax))^* = \sup$$

$$g(x) = f(Ax)$$

$$\sup_x \{y^T x - f(Ax)\}$$

$$= \sup_x \{y^T A^T Ax - f(Ax)\}$$

$$= \sup_{Ax} \{(A^{-T} y)^T x - f(x)\} = f^*(A^{-T} y)$$